

PROBLEM STATEMENT

Nested Logarithm Chain Rule

Find the derivative with respect to x of the function:

$$y = \log(\log(\log x))$$

STEP 1 প্রদত্ত ত্রি-স্তর যৌগিক লগারিদম ফাংশনটি উল্লেখ করি (Given Triple Nested Function)

$$y = \log(\log(\log x))$$

STEP 2 উভয়পক্ষে x -এর সাপেক্ষে অন্তরীকরণ অপারেটর প্রয়োগ করি (Applying Differentiation Operator w.r.t. x)

$$\frac{dy}{dx} = \frac{d}{dx} [\log(\log(\log x))]$$

STEP 3 সর্ববহিঃস্থ লগারিদমের জন্য চেইন রুল প্রয়োগ করি (Differentiating Outermost Logarithm)

$$\frac{dy}{dx} = \frac{1}{\log(\log x)} \cdot \frac{d}{dx} [\log(\log x)]$$

STEP 4 মধ্যবর্তী লগারিদমের জন্য চেইন রুল প্রসারিত করি (Differentiating Intermediate Logarithm)

$$\frac{dy}{dx} = \frac{1}{\log(\log x)} \cdot \frac{1}{\log x} \cdot \frac{d}{dx} (\log x)$$

STEP 5 সর্বঅভ্যন্তরীণ $\log x$ -এর অন্তরজ নির্ণয় করে গুণফল আকারে সাজাই (Differentiating Innermost Logarithm)

$$\frac{dy}{dx} = \frac{1}{\log(\log x)} \cdot \frac{1}{\log x} \cdot \frac{1}{x} = \frac{1}{x \log x \log(\log x)}$$

FINAL ANSWER
$$\frac{dy}{dx} = \frac{1}{x \log x \log(\log x)}$$

BOX 1: MATHEMATICAL FOUNDATIONS & FORMULA REFERENCE

- Triple Chain Rule Layering:** Let $u = \log(\log x)$ and $v = \log x$. Then $y = \log u$.
$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx} = \frac{1}{u} \cdot \frac{1}{v} \cdot \frac{1}{x}$$
- Standard Logarithmic Derivative:** $\frac{d}{dt}[\log t] = \frac{1}{t}$.
- General n -fold Nested Logarithm:** For $y = \log^{[n]}(x)$, $\frac{dy}{dx} = \frac{1}{x \cdot \log x \cdot \log(\log x) \cdots \log^{[n-1]}(x)}$.

BOX 2: COMMON STUDENT TRAPS & PITFALLS

- Nested Logarithm vs. Power of Logarithm:** $\log(\log(\log x)) \neq (\log x)^3$. The derivative of $(\log x)^3$ is $\frac{3(\log x)^2}{x}$, which is entirely different!
- Dropping Intermediate Factors:** Jumping straight from the outer log to $\frac{1}{x}$ without including $\frac{1}{\log x}$.
- Order of Factors in Denominator:** Conventional standard is writing in ascending complexity: $x \log x \log(\log x)$.

BOX 3: REAL DOMAIN & DIFFERENTIABILITY ANALYSIS

- Innermost condition:** $x > 0$.
- Middle condition:** $\log x > 0 \implies x > 1$.
- Outermost condition:** $\log(\log x) > 0 \implies \log x > 1 \implies x > e \approx 2.71828$.
- Strict Real Domain:** Domain = (e, ∞) .
- Differentiability:** Infinitely differentiable and monotonically increasing on (e, ∞) .

BOX 4: EXPLICIT SUBSTITUTION METHOD (ধাপে ধাপে চলক প্রতিস্থাপন)

- Let $u = \log x \implies \frac{du}{dx} = \frac{1}{x}$
- Let $v = \log u \implies \frac{dv}{du} = \frac{1}{u} = \frac{1}{\log x}$
- Then $y = \log v \implies \frac{dy}{dv} = \frac{1}{v} = \frac{1}{\log(\log x)}$
- Multiplying the three rates of change:
$$\frac{dy}{dx} = \frac{dy}{dv} \cdot \frac{dv}{du} \cdot \frac{du}{dx} = \frac{1}{\log(\log x)} \cdot \frac{1}{\log x} \cdot \frac{1}{x} = \frac{1}{x \log x \log(\log x)}$$

BOX 5: EXAM MARKS BREAKDOWN (AMIE SECTION-A FORMAT)

- Step 1–2 (Function Setup & Chain Rule Strategy):** 1.0 Mark
- Step 3 (Outermost Layer Derivative):** 1.0 Mark
- Step 4–5 (Intermediate & Innermost Layer Derivatives):** 1.5 Marks
- Final Expressive Form & Answer:** 0.5 Mark
- Total Allocation:** 4.0 Marks